

Setup Guide

Quantization & Speculative Decoding Benchmarking Engine Setup Guide

1. Prerequisites

- Python 3.10+ and pip (or uv).
- For LIVE benchmark runs: a CUDA GPU and the optional requirements-live.txt stack (torch, transformers, bitsandbytes, autoawq, gptqmodel). Simulated runs need no GPU.

2. One-command setup

```
python3 infra/setup.py
```

Creates .venv and installs the pinned requirements.txt.

3. Run the full pipeline (simulated, deterministic)

```
python -m orchestrator.pipeline --mode simulate --seed 42
```

Outputs:

- artifacts/rawmetrics.json merged metrics with provenance
- artifacts/paretofrontier.png Pareto efficiency frontier chart
- artifacts/technicalwriteup.md technical writeup
- results/runstatus.json pipeline status report

4. Run individual modules

```
python -m benchmarks.quantizationrunner --mode simulate --seed 42
```

```
python -m benchmarks.speculativerunner --mode simulate --seed 42
```

```
python scripts/generatepareto.py
```

```
python scripts/generatewriteup.py
```

5. Run on real hardware (LIVE mode)

```
pip install -r requirements-live.txt # GPU host only
```

```
python -m orchestrator.pipeline --mode live
```

6. View the docs / landing page locally

```
bun server.js (or: npm run dev)
```

Serves ./web and ./artifacts at http://0.0.0.0:4173

7. Push to GitHub (gated DevOps task)

```
sh scripts/pushtogithub.sh
```

Runs the pre-push checklist, commits, and fast-forward pushes to origin.